Submitted By:

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Assignment:

CEP-1

Submitted To:

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For the fulfillment of,

Object-Oriented Programming Lab

Department of Computer Engineering

University of Engineering and Technology, Lahore

Pac-Man Game Lab Report:-

1. Introduction

For this lab, I had to create a basic version of the classic Pac-Man game that runs right in the console.The game lets you move Pac-Man around a maze, eat dots to get points, and avoid two ghosts. You start with 3 lives, and the goal is to eat every dot without getting caught. If a ghost touches you, you lose a life and everything resets. Run out of lives and it's game over.

2. What the Program Does

The main goal was to build a playable game that shows:

- A fixed maze with walls and dots
- Player movement using keyboard (W/A/S/D)
- Random ghost movement
- Score tracking
- Lives system
- Collision detection
- Win and lose conditions

3. How the Code Is Structured

Here's a quick look at the main classes I used:

a. **Map.java:**
This holds the whole 9-row by 12-column maze. It stores walls (#), dots (.), and empty spaces. It also has methods to check if a spot is a wall, eat a dot, print the map, and tell when all dots are gone.

b. **PacMan.java:**
Controls where Pac-Man is and lets him move. He only moves if there's no wall in the way, and if he lands on a dot he eats it and adds 10 points.

c. **Ghost.java:**
There are two ghosts. They pick a random direction each turn and move only if it's not a wall. They start at fixed spots.

d. **Position.java:**
A simple helper class that just keeps track of any (x, y) spot on the map and can compare two positions to see if they're the same.

e. **Score.java and Lives.java:**
one adds up points, the other counts down your 3 lives and checks if the game is over.

f. **Gameboard.java:**
This is the "boss" class with the main method. It sets everything up, runs the game loop, prints the map every turn, reads your keyboard input, moves Pac-Man and the ghosts, checks for collisions, and handles winning or losing.

4. How the Game Loop Works

The game runs in a while loop inside Gameboard.java:

1. Print the current map (shows "P" for you and "G" for ghosts).

2. Show your current score and lives.

3. Ask for your move (W/A/S/D or Q to quit).

4. Move Pac-Man.

5. Move both ghosts randomly.

6. Check if any ghost is on the same spot as Pac-Man. If yes, lose a life, reset positions, and check if you're out of lives.

7. If you ate every dot, you win!

5. Testing and Results

I compiled and ran the program a bunch of times to test it.

- Moving around worked great. You can't walk through walls.

- Eating dots added exactly 10 points each time and the dots disappeared.

- Ghosts moved in random directions but never went through walls.

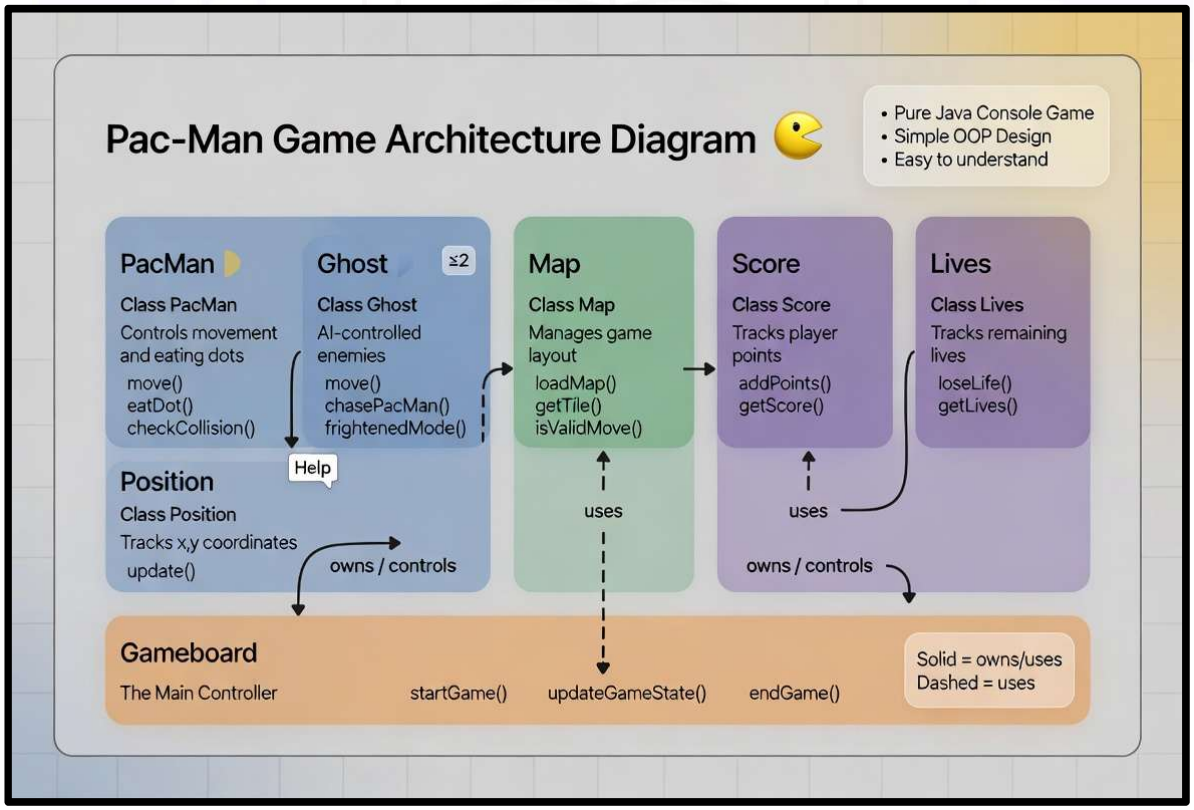
- When a ghost caught me, it printed "GHOST CAUGHT YOU!", took away one life, and put everyone back to starting spots.

- After eating all dots, it showed the win message with my final score.

- If I lost all 3 lives, it correctly ended the game.

- Pressing Q let me quit early and still showed my score.

Conclusion

Overall, the Pac-Man game turned out better than I expected. It meets all the requirements, runs without bugs, and is actually enjoyable to play. I'm proud of how the different classes work together like a team. This project made OOP concepts click for me way more than just doing simple exercises.

Architecture Diagram:-